Rajveer Gupta

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Summary

Developer who builds **reliable, privacy-first systems**: camera-fault detection that ignores glitches, fraud checks that avoid false alarms, and face recognition that never stores raw images. Looking for **AI/ML, SDE or AI security internships**.

Education

Indian Institute of Information Technology, Nagpur Aug 2024 – Jun 2028
B.Tech, Computer Science & Engineering — CGPA: 9.08/10 *Nagpur, India*
Coursework: OOP (AA), Software Engineering (AA), Operating Systems (AB), Programming Language Design (AB), DSA, DBMS

Achievements

- **AIR 471 of 32K teams, Amazon ML Challenge 2026** — team lead; matched **11.7M records** to real businesses.
- **Amazon ML Summer School 2026** — selected among the top 3,000 of 134,000+ applicants.
- **Best Innovation Award**, Tech Expo 2026, IIT Hyderabad • **Microsoft Imagine Cup 2025** — MVP Round.
- Finalist, IEEE IES Generative AI Hackathon 2026 • 2nd Place, Analytica ML/DL Hackathon, IIIT Nagpur.

Experience

Research Intern, Computer Vision & Deep Learning — IIIT Nagpur (Dr. Nishat Afshan Ansari) 2026
What it does: spots which factory camera has gone bad (blurred, noisy, blocked or dead) and rebuilds its view from nearby cameras.

- **Detect:** monitors every feed frame by frame; **hysteresis-based fault classification** flags a camera only when the fault persists, so one glitchy frame does not raise a false alarm.
- **Reconstruct:** rebuilds a dead camera's view from neighbors via **3D Gaussian reconstruction**; diffusion fills the gaps.
- **Restore:** cleans blurred or noisy frames with a Mamba-CNN network that runs in **linear, not quadratic, time**.
- **Results:** 30.02 dB PSNR / 0.893 SSIM (Wildtrack) and 32.42 dB / 0.953 SSIM (GoPro), beating NAFNet and MoCE-IR.

Projects

ReturnGuard — Return-Fraud Detection Service for E-Commerce 2026
GitHub / Demo — Python, Flask, REST APIs, SQLite, LangGraph, DINOv2, Gemini VLM
What it does: checks whether a returned product is really the one that was delivered, catching swaps or damage before the refund.
Features: delivery-vs-return photo check, automatic *accept / human review / reject*, risk score with written reasoning.

- **Architected the complete system**: Flask REST API, SQLite store, and a LangGraph workflow with conditional routing and a **reusable VLM subgraph** run **in parallel** per photo view (fan-out / fan-in) before an LLM judge.
- **Cut cost with a two-tier pipeline**: DINOv2 embeddings auto-clear near-identical items; VLMs review the rest.
- Assigned every case a **0–100 risk score**, logged with its reasoning and verdict via **parameterized SQL queries**.
- **Tuned for precision so honest customers are not refused:** found DINOv2 rated *different* same-material products 85–92% similar and tightened the threshold; added blur-aware weighting. **0.93 precision / 0.88 F1**.

Memora — Privacy-First Memory-Care Assistant for Alzheimer's Patients 2025 – 2026
GitHub / Live / Demo / Case Study / Figma — Python, DeepFace (ArcFace), BERT
What it does: helps Alzheimer's patients recognize family. **Four features:** (1) point the phone at someone to hear who they are in a loved one's voice; (2) safe zones with SOS and live location if the patient wanders; (3) finds misplaced objects; (4) detects distress.

- Led a 4-service team project (React, Node.js, FastAPI, Supabase) and **built the ML layer**: ArcFace face recognition (**98.5% on LFW**, cosine threshold 0.68) and a BERT speech-distress model.
- Protected **user privacy**: frames are converted to 512-d embeddings, then deleted; **no raw face image is stored**.

Stock Market Simulator — Object-Oriented Trading Engine in C++ 2025
GitHub — C++, STL, OOP, nlohmann/json, Python, Flask
What it does: a browser stock-market game where traders buy and sell shares, plan SIPs and compete on a leaderboard.

- Developed a C++ engine of 7 classes (Trader, Company, Stock, Transaction, SIP, Market, Leaderboard) using encapsulation, inheritance, friend classes for controlled access, and a non-instantiable static `Market` utility.
- Stored holdings in `unordered_map` (O(1) lookup) with JSON persistence; a Flask layer drives the C++ binary via pipes.

Publications

A. Gupta, A. Singh, **R. Gupta**, P. Jain, T. Patidar, "Fault Tolerant Industrial Vision Framework Through Semantic-Guided Cross-View Temporal Reconstruction and Restoration," *IEEE IRAI 2026*, Melbourne (presented); 32.42 dB PSNR.

Skills

Languages: C++, C, Python, Java, SQL
Core CS: DSA (250+ problems, LeetCode / GfG), OOP, OS, DBMS, Computer Networks (basics), System Design
Backend & Tools: Flask, FastAPI, REST APIs, SQLite, PostgreSQL, MySQL, Docker, Git, Linux
ML & Deep Learning: PyTorch, scikit-learn, XGBoost, LightGBM, CNNs, Transformers, ViT, BERT, Diffusion, RL, OpenCV
LLMs & Agents: LangGraph, LangChain, RAG, Hugging Face, PEFT (LoRA / QLoRA), VLMs, DINOv2, FAISS, ChromaDB